Photocrosslinkable Materials for the Fabrication of Tissue-Engineered Constructs by Stereolithography

Rúben F. Pereira and Paulo J. Bártolo

Abstract Stereolithography is an additive technique that produces three-dimensional (3D) solid objects using a multi-layer procedure through the selective photo-initiated curing reaction of a liquid photosensitive material. Stereolithographic processes have been widely employed in Tissue Engineering for the fabrication of temporary constructs, using natural and synthetic polymers, and polymer-ceramic composites. These processes allow the fabrication of complex structures with a high accuracy and precision at physiological temperatures, incorporating cells and growth factors without significant damage or denaturation. Despite recent advances on the development of novel biomaterials and biocompatible crosslinking agents, the main limitation of these techniques are the lack number of available photocrosslinkable materials, exhibiting appropriate biocompatibility and biodegradability. This chapter gives an overview of the current state-of-art of materials and stereolithographic techniques to produce constructs for tissue regeneration, outlining challenges for future research.

1 Introduction

Tissue engineering is recognized as a promising field to overcome some of the limitations of existing clinical treatments for the repair of damaged and dysfunctional tissues or organs, such as shortage of donors, chronic rejection or transmission of diseases. This interdisciplinary field involves principles from biological sciences and engineering for the development of biological substitutes to restore, maintain, or improve tissue function [9, 11, 15]. Despite the recent advances on the interaction between living cells and materials [32, 36, 112] allowed the development of functional scaffolds to support cell activity and potentially promote the repair of different tissues like skin [21, 84], bone [51, 110] or cartilage [47, 104], the development of cost-effective approaches for the regeneration of damaged tissues remains a great challenge.

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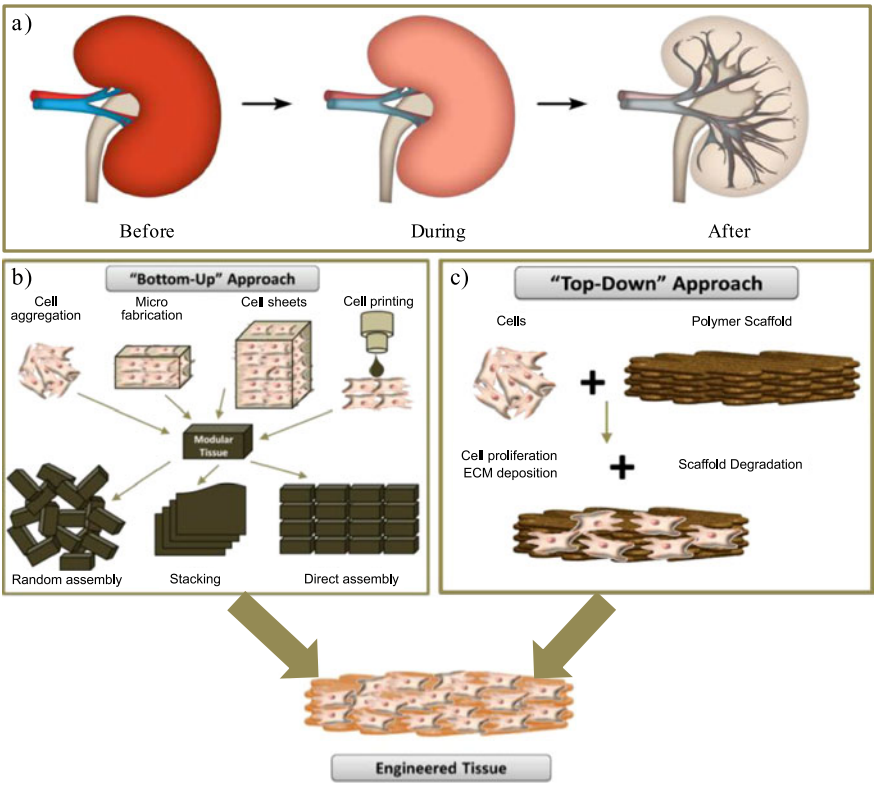


Fig. 1 Representative image of a kidney at different stages of the decellularization process (**a**). Bottom up (**b**) and top-down (**c**) approaches for tissue engineering (adapted from [75])

Decellularized tissue matrices gained increasing attention as a template for organ regeneration. These matrices, shown in Fig. 1a, were obtained from a donor and its cellular components removed, this way avoiding the risk of rejection and maintaining the 3D structure of the extracellular matrix (ECM) as a scaffold for organ regeneration [78, 88]. Currently, two fundamental strategies are considered for the development of biological substitutes for damaged tissues, namely the bottom-up (Fig. 1b) and the top-down (Fig. 1c) approaches [15, 75].

The bottom-up approach has emerged as a new method for the development of 3D biomimetic substitutes exploring the self-ability of cells to synthesize their own ECM without the need of supporting materials [15, 69, 73, 88]. Complex structures can be obtained through this approach by assembling modular tissues produced by different techniques, such as self-assembled aggregation, microfabrication of cell-laden hydrogels or direct printing [25, 75]. The top-down approach, commonly used one, is based on the use of porous and biodegradable matrices (scaffolds) to support cell attachment, proliferation and differentiation, guiding the formation of new tissue in an organized way [14, 15]. This approach involves the fabrication

Table 1 Biological, physical and mechanical requirements of scaffolds [1, 8, 14, 15]

Biological requisites	
Biocompatibility	Scaffold must be non-toxic and interact with biological tissues without inducing adverse responses
Biodegradability	Scaffold material should be gradually degraded in non-toxic products with appropriated molecular weights to allow the clearance from the human body
Degradation rate	The degradation rate should be adjustable and match the regeneration rate of the new tissue
Porosity	Scaffold must exhibit adequate pore size and interconnected pores. These properties are fundamental to promote an efficient cell seeding, nutrient and waste exchange, vascularization and tissue in-growth
Bioactivity	Scaffolds should be able to stimulate the attachment, proliferation and differentiation of the seeded cells, guiding the growth of the new tissue. The constructs should also be able to incorporate and deliver drugs and growth factors according to specific release profiles
Mechanical and physical requisites	
Mechanical strength	Scaffolds must present adequate strength and stiffness to support stresses in the host tissue environment. Mechanical properties should be similar to those in the native tissue, providing a temporary support to the tissue formation. During regeneration, scaffolds gradually transfer the mechanical loads and stresses to the new tissue
Surface finish	The surface chemistry of scaffolds should promote an optimal biomechanical coupling between the scaffold and the tissue, promoting cell attachment, differentiation and proliferation
Sterilization	Scaffolds must be easily sterilized using thermal, chemical or radiation processes, without degradation or modification of the material properties

of 3D biocompatible scaffolds and the manipulation of living cells and signalling molecules (e.g. growth factors), as well the *in vitro* culture of the cellular tissue-engineered constructs within bioreactors to promote the growing of clinically relevant healthy tissues. Scaffolds, from either natural or synthetic materials, play an important role on tissue regeneration, mimicking the function of the natural EMC of the human body. These structures act as a temporary support for the seeded autologous or allogeneic cells proliferation, differentiation, and synthesizing their own ECM [14, 15, 69]. Scaffolds need to successfully satisfy several requirements, as indicated in Table 1.

Several techniques can be used to produce scaffolds for tissue engineering applications, which can be classified as non-additive and additive biomanufacturing techniques.

Conventional techniques like solvent casting, freeze-drying, phase separation, gas foaming, melt molding and particle-leaching, are used to produce 3D scaffolds with relative control over the micro- and macro-scale features [41, 46, 77, 95, 106, 108]. However, these techniques present several limitations, such as the lack of control over the pore size and interconnectivity, porosity and pore spatial distribution [1, 15, 82, 109], leading to an inadequate vascularization and heterogeneous distribution of cells, promoting a non-uniform tissue growth [68]. In addition, these techniques usually employ toxic organic solvents, which prevent the incorporation of cells and other biological molecules during fabrication [15, 82].

Biomanufacturing represents a group of non-conventional fabrication techniques for the production of biological constructs for tissue engineering applications through the use of additive technologies, biodegradable and biocompatible materials, cells and growth factors [6, 7]. Additive biomanufacturing techniques produce complex 3D scaffolds in a layer-by-layer pattern from a CAD model, providing precise control over the pore size, porosity and pore interconnectivity [8, 15, 18, 82]. Several techniques, such as vat photopolymerization or stereolithographic processes, powder bed fusion processes, extrusion-based processes and inkjet processes have been developed, processing a wide range of materials, with a high level of automation and reproducibility [69]. Among these techniques, the vat photopolymerization processes are widely used for the production of scaffolds both containing or not encapsulated cells and growth factors, due to the ability to induce curing at physiological temperatures with a high accuracy, precision and resolution. Stereolithographic processes and different materials have been successfully used to produce tissue-engineered constructs. These materials are described in this chapter, highlighting different photofabrication approaches and applications.

2 Stereolithography

Stereolithography produces 3D solid objects in a multi-layer procedure through the selective photo-initiated curing reaction of a liquid photosensitive material containing a low-molecular weight pre-polymer, additives and photo-initiators [8, 15, 17, 65, 74]. The curing reaction is induced by a light source like ultraviolet (UV), infrared (IR) or visible light, supplying the necessary energy to bond a large number of small molecules, forming a highly cross-linked polymer [13, 15]. Radical polymerization is the most commonly used method to allow the photopolymerization, involving the generation of reactive species (free radicals) by the interaction with the incident light [26]. These species induce a curing reaction forming an insoluble and cross-linked 3D network. During this process, the liquid polymeric solution increases its viscosity as a result of the gelation, forming an elastic sol-gel structure. The sol fraction decreases and the polymer becomes more viscous, due to an increase in the number of cross-links between the polymeric chains. As the

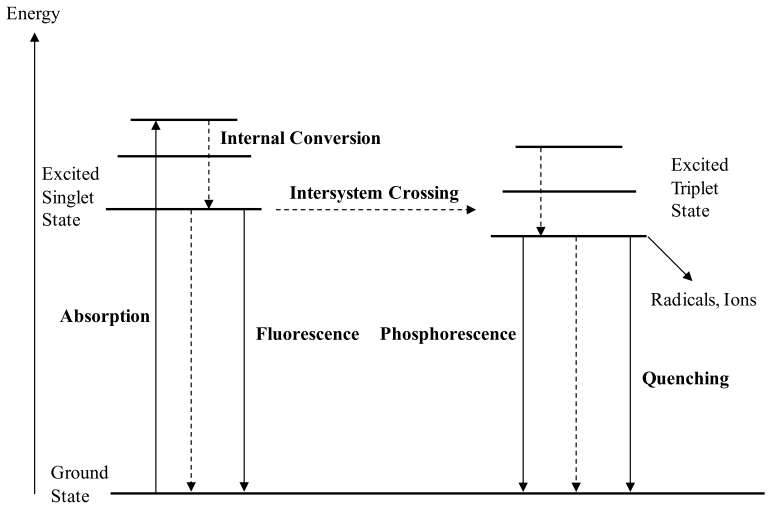


Fig. 2 Jablonski energy diagram

reaction proceeds, the material becomes more cross-linked, its molecular weight increases and a glassy solid material is formed [13]. The curing reaction is highly dependent on the light intensity, temperature, irradiation time, and photo-initiator concentration [13]. Either a single-photon polymerization or a two-photon polymerization (2PP) can be used to induce the curing reaction. The chemical principle of these two processes is similar, though they differ in the number of absorbed photons required to induce the polymerization process [13, 82]. When a molecule absorbs light, the electrons are set into motion by the oscillating electric field, which is promoted from the highest occupied molecular orbital to an unoccupied molecular orbital with the formation of an excited singlet state molecule. However, this excited molecule is a short living species (less than 10^{-8} s), which disappears by various competitive processes dissipating the excited energy. The absorption of light by a molecule and the subsequent evolution of its excited states can be observed in Fig. 2 through the Jablonski energy diagram.

Two processes can be identified, the photophysical and the photochemical ones, as follows:

- Photophysical processes:
  - Radiative
  - Non-radiative
- Photochemical processes.

Radiative mechanisms involve both the absorption of a photon or more by a molecule in its ground state (S_0) and the emission of energy from an electronically excited state, by either fluorescence (de-excitation of an excited state with the same spin multiplicity as the ground state) or phosphorescence (de-excitation of

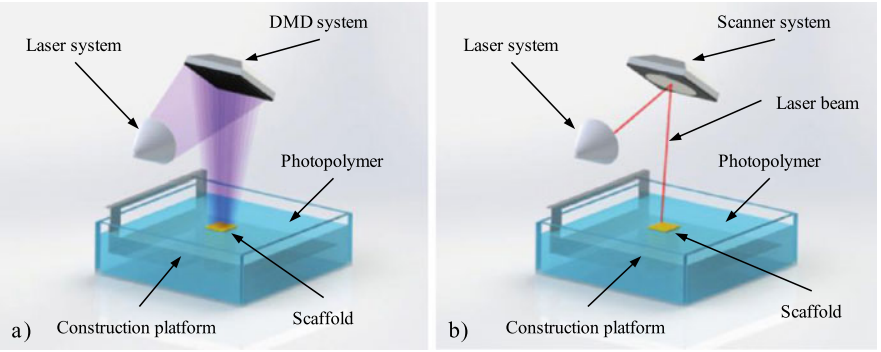


Fig. 3 Stereolithographic processes: mask-based (**a**) and laser writing (**b**) methods

an excited state with different spin multiplicity as the ground state). Non-radiative processes include internal conversion (IC) and intersystem crossing (ISC), which occurs among different spin multiplicity states creating excited triplet states (T_1). The photochemical processes imply the transformation of the starting molecule through cleavage processes, electron transfer reactions, hydrogen abstraction, etc. Most photochemical reactions occur only via excited triplet states, which are longer live species (greater than 10^{-6} s). Singlet or triplet states are electronic states where the molecule has either paired electrons or unpaired electrons, respectively.

Stereolithographic processes usually employ two distinct methods of irradiation: the mask-based method (Fig. 3a) and the direct or laser writing method (Fig. 3b) [12].

In the mask-based method, an image is transferred to a liquid polymer by irradiation through a patterned mask, which contains transparent zones corresponding to the sections of the model to be built [8]. This process enables the curing of one entire layer in just one irradiation step, this way reducing the fabrication time and avoiding undesirable polymerizations as a result of the low-density flux of light over the polymer surface [15, 71, 85]. However, the mask-based approach requires the generation of a great number of masks with precise alignments, which can be done by using Liquid Crystal Display (LCD) panels and Digital Micromirror Devices (DMD's) as dynamic pattern generators [13, 85].

The direct or laser writing method is the most commonly used one, involving the use of a focused laser beam to selectively irradiate and solidify the liquid photopolymer [8]. In this case, the stereolithography apparatus consists of a computer, a vat containing a photosensitive liquid polymer, a moveable platform in which the model is built, a laser to irradiate and cure the photosensitive resin, and a dynamic mirror system to guide and project the laser beam over the polymer surface. After the curing of the first layer, the platform dips into the polymer vat and leaves a thin film of liquid polymer in the surface of the first layer, which is then irradiated to produce the second layer [15, 16]. Once the depth of curing is larger than the resin layer (Fig. 4), a good adhesion between the different layers of the model is ensured [71].

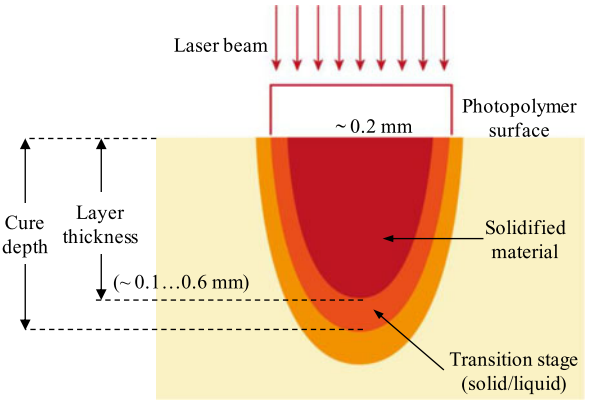


Fig. 4 Irradiation phenomena

Conventional stereolithography has been successfully applied for the fabrication of tissue-engineered constructs using several natural and synthetic materials [58, 68, 115]. However, the fabrication of porous 3D constructs with an interconnected micro-architecture is limited by the resolution of the system [27]. Microstereolithography is a process that evolved from conventional stereolithography, allowing the fabrication of complex 3D constructs with a micro-scale resolution of about 200 nm [27, 55, 71]. In this process, the laser beam is more precisely focused, reducing the spot size to a few micrometers in diameter, this way enhancing the resolution of the system [13, 56]. Today, there is a great interest in the use of 2PP processes for the fabrication of various devices, such as microneedles, sensors and scaffolds for different biological and biomedical applications [39, 48, 53]. These processes are able to produce 3D structures with submicron resolution, enabling an ultra-fast fabrication process at greater depth [8]. However, 2PP systems are very expensive, operating with a single material type that prevents the fabrication of multimaterial constructs [85]. These equipments usually employ a femtosecond titanium:sapphire laser (short pulse width and high peak power) without photo-masks, operating at approximately 800 nm wavelength to induce the polymerization [53, 74, 82]. A great variety of natural and synthetic photosensitive materials can be used for the 2PP technique, like acrylate-based polymer, zirconium sol-gels, organically modified ceramic materials, aliphatic polyesters and gelatin [52, 74, 79, 105].

Despite these processes enable to produce complex 3D scaffolds with pre-defined micro/nanoscale architectures, the major limitation of stereolithographic processes for tissue engineering applications consists on the lack of available photocurable, biocompatible and biodegradable materials.

3 Materials for Stereolithography

The first polymeric systems developed for stereolithography were based on low-molecular weight polyacrylate or epoxy macromers, which present a rapid cure and

enable an easy modification at the ester functionality [13, 44, 71, 91]. However, the obtained structures are scarcely applied on tissue engineering as they are predominantly glassy, rigid and brittle, preventing the fabrication of flexible and elastomeric structures to resemble the properties of human tissues [13, 58]. In addition, these polymeric systems are usually not biocompatible or biodegradable, thereby preventing their application for the fabrication of scaffolds [35, 44, 91]. To address the need for photocurable biomaterials, both biocompatible and biodegradable, significant technological advances are used to develop novel polymeric systems. The fabrication of scaffolds through stereolithography involves the use of natural and synthetic polymers, and polymer/ceramic blends.

3.1 Polymers

Polymeric materials are the most commonly used materials for biomedical applications, due to the wide range of its properties, easy processing and versatility. Both natural and synthetic polymers can be modified to allow the processing through stereolithography, without affecting or even improving the interaction with living cells.

Biocompatible hydrogels, based on either natural or synthetic polymers, represent a relevant group of materials widely employed for several biomedical applications, including tissue engineering [97], wound dressings [84], controlled drug delivery [113] and cell encapsulation [103]. Hydrogels are tridimensional hydrophilic networks with the ability to absorb and retain large amounts of water without dissolution [84], due to the establishment of physical (reversible) or chemical (irreversible) bonds between the polymeric chains [43, 69, 98, 100]. These are attractive materials for tissue engineering applications due to its excellent biocompatibility, biodegradability, elasticity, smoothness and compositional similarities regarding the ECM of the human body [43, 98, 101]. In addition, some hydrogels can be photopolymerized using *in vitro* and *in vivo* conditions in the presence of cells and photoinitiators [13, 63, 100, 111], which can potentially improve its use for the *in situ* regeneration of damaged tissues. The high water content of the hydrogels makes them useful carriers for the transport and delivery of fragile molecules (e.g. proteins, drugs or cells), providing a 3D environment similar to the one in human tissues, protecting these molecules from denaturation or degradation [69, 98, 101]. It is possible to manipulate the diffusion and transport of biological materials by changing both the density of cross-links and the pore size of the gel network. The control over the cross-links density also enables to tailor different properties, such as its swelling behavior, degradation rate, mechanical properties, pore size and permeability [50, 69].

Hydrogels used in stereolithography comprise natural polymers (e.g. alginate, chitosan, hyaluronic acid, gelatin), synthetic polymers (e.g. poly(ethyleneglycol) (PEG), propylene fumarate (PPF), poly(ε-caprolactone) (PCL)) and a combination of both [13, 59, 100, 101]. Synthetic and natural hydrogels are usually modified using photoreactive and crosslinkable groups, such as acrylates and methacrylates, to enable their processing by stereolithographic processes [4, 69, 71, 111].

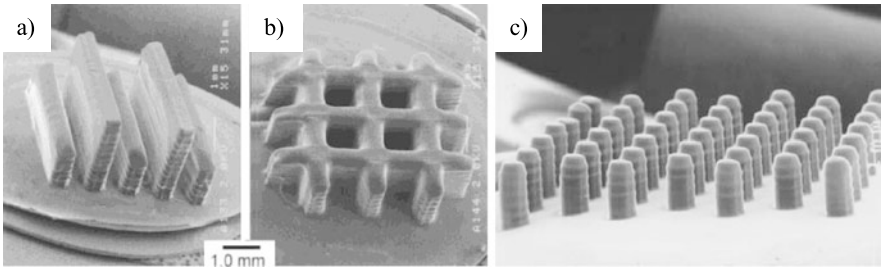


Fig. 5 Scanning electron microscopy (SEM) images of microbanks (**a**), microwells (**b**) and microneedle-array (**c**) built by stereolithography using the CL and TMC oligomers [66]

3.1.1 Synthetic Polymers and Hydrogels

The interest on synthetic polymers concerns good and adjustable physical, chemical and mechanical properties, easy processing into a wide range of shapes and large-scale production. On the other hand, their main drawback regard its limited biocompatibility, poor interaction with living cells, and in some cases the toxicity of the degradation products [87, 96]. The surface of synthetic polymers can be modified through different approaches, such as plasma treatment, polymer coating, chemical modification, peptide immobilization and photochemical modifications, to improve cell interactions or provide specific interactions with different cell types [40, 102].

In stereolithography, the most used biodegradable macromers are based on functionalized oligomers containing hydrolyzable ester- or carbonate linkages in the main chain [71]. Examples include PPF [56], PCL [35], poly(D,L-lactide) (PDLLA) [70] and poly(trimethylene carbonate) (PTMC) [89]. Usually, these polymers are mixed with reactive and non-reactive diluents, such as diethyl fumarate (DEF), to control the viscosity and the degree of cross-linking, allowing the fabrication of constructs with adequate mechanical properties [35, 50, 58, 71].

Matsuda et al. [67] reported the preparation of photocurable liquid biodegradable copolymers through the ring-opening copolymerization of ε-caprolactone (CL) and trimethylene carbonate (TMC), using polyol as initiator and tin(II) 2-ethylhexanoate as a catalyst. The copolymers were subsequently derivated at the hydroxyl end with a photodimerizable coumarin group. Results showed that higher coumarin functionality and UV light intensity, and a reduced layer thickness of the liquid film precursor increases the photocuring reaction. The same oligomers were used by Matsuda and Mizutani [66] to produce photocurable copolymers, using trimethylene glycol or PEG as initiator and an acrylate group (acryloyl chloride) to end-functionalize the oligomers. Different structures were produced using the photocurable copolymers through stereolithography, as shown in Fig. 5.

PPF, an unsaturated linear polyester, has been used in stereolithography due to its ability to be cross-linked through the carbon-carbon double bonds, presenting excellent mechanical properties [54, 87]. The subunits of PPF can be crosslinked by using different agents, such as DEF, methyl methacrylate and N-vinylpyrrolidone [54]. In addition, PPF also undergoes degradation into biocompatible and non-toxic

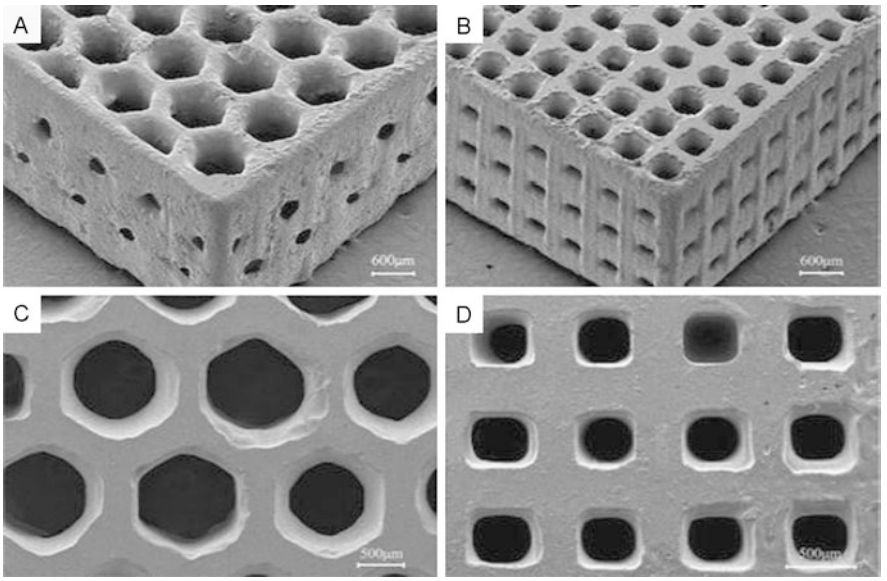


Fig. 6 SEM images of PPF scaffolds, with different pores shapes (hexagon (**A**, **C**) and square (**B**, **D**) pores) and pore sizes (600 μm for hexagon and 508 μm for square) [58]

fumaric acid and propylene glycol products by simple hydrolysis of the ester bonds [57, 58, 87].

Lee et al. [58] produced 3D scaffolds for tissue engineering using a UV curable polymer solution, consisting of PPF, DEF as a solvent and bisacrylphosphrine oxide (BAPO) as a photoinitiator. The composition of the photocurable solution and the laser parameters were optimized and scaffolds with different pore sizes, pore shapes and porosities were produced (Fig. 6). Similarly, Lee et al. [57] used a microstereolithography system to produce PPF scaffolds with control over the pore size, porosity, interconnectivity and pore distribution. After 1 week of *in vitro* culture, fibroblasts showed good adhesion and spreading on both bottom and side construct walls.

However, synthetic polymers like PPF are hydrophobic and low bioactive materials. To solve these limitations, Lan et al. [54] coated PPF scaffolds produced by microstereolithography with accelerated biomimetic apatite and arginini-glycine-aspartic acid peptide coating. Similarly, Shin et al. [94] used three different peptides, Arg–Gly–Asp (RGD), cyclo RGD and a mixture of RGD-KRSR (lysine–arginine–serine–arginine), to improve the surface properties of PPF/DEF scaffolds for bone tissue engineering applications. The modified scaffolds were seeded with MC3T3-E1 pre-osteoblasts, and the effect of each peptide on the cell adhesion, proliferation and differentiation was evaluated. Results showed that the peptide modification enhanced the adhesion and proliferation of MC3T3-E1 pre-osteoblasts, comparatively to the non-modified scaffolds.

Stereolithography has also been used to produce scaffolds for the delivery of growth factors. Lee et al. [56] produced 3D scaffolds containing bone morphogenetic protein-2 (BMP-2)-loaded poly(DL-lactic-coglycolic acid) (PLGA) microspheres (Fig. 7), by the polymerization of a suspension consisting of a PPF/DEF photopolymer and microspheres. In this work, scaffolds were also produced through a conventional process (particulate leaching/gas foaming) to evaluate the influence of the fabrication process on the scaffold performance. Results showed that scaffolds, produced by microstereolithography, provide a better environment for cell proliferation and differentiation. To evaluate the *in vivo* bone formation, scaffolds were implanted into a rat cranial defect. After 11 weeks of implantation, it was possible to observe a significant bone formation on the defect treated using the BMP-2-loaded scaffold, produced by microstereolithography (Fig. 7).

Thermoplastic aliphatic polyesters, such as PCL, poly(lactic acid) (PLA) and poly(glycolic acid) (PGA), represent another class of synthetic polymers extensively applied for the fabrication of tissue engineering scaffolds through stereolithography. The large number of existing aliphatic polyesters offers the possibility to prepare structures with distinct properties. For example, lactide-based precursors have been used to fabricate hard and rigid structures for both orthopedic and bone tissue applications, while the copolyester precursors are employed for the fabrication of flexible and elastomeric structures suitable for soft-tissue applications [92].

PCL is a biodegradable, biocompatible and semi-crystalline polymer FDA approved for various applications, such as sutures, wound dressings and stents [28, 83, 99]. This material presents a low melting point and its degradation kinetics, physical and mechanical properties can be easily adjusted by different approaches, including the (i) manipulation of the polymer molecular weight, (ii) the copolymer ratio, (iii) the blending with other polymers, and (iv) the incorporation of labile bonds into the backbone [35, 87].

Elomaa et al. [35] synthesized three-armed PCL oligomers by ring-opening polymerization of ε-caprolactone monomers. The photocrosslinkable PCL-based resin was end-functionalized with methacrylic anhydride, and subsequently employed to produce 3D porous scaffolds trough a mask stereolithographic system. The produced scaffolds exhibited porosity of 70.5 %, pore size in the range of 400–500 μm, and a high interconnectivity between pores without material shrinkage. NIH3T3 fibroblasts, cultured on photocrosslinked PCL networks, can be easily attached presenting uniform spreading.

Two-photon polymerization has been explored to produce scaffolds using PCL-based polymers. Claeyssens et al. [31] fabricated 3D structures composed of the biodegradable triblock copolymer poly(ε-caprolactone-co-trimethylenecarbonate)-b-poly(ethylene glycol)-b-poly(ε-caprolactoneco-trimethylenecarbonate), using 4,4′-bis(diethylamino) benzophenone as the photoinitiator. Constructs with different geometries were prepared with a resolution of 4 μm (Fig. 8). Fibroblasts, cultured onto spin-coated thin films after photopolymerization, remained viable and showed comparable cell attachment and division regarding cells cultured on glass surfaces (control), which indicates that the developed material do not affect cell proliferation. In a similar work, Koskela et al. [53] used the 2PP technique to produce

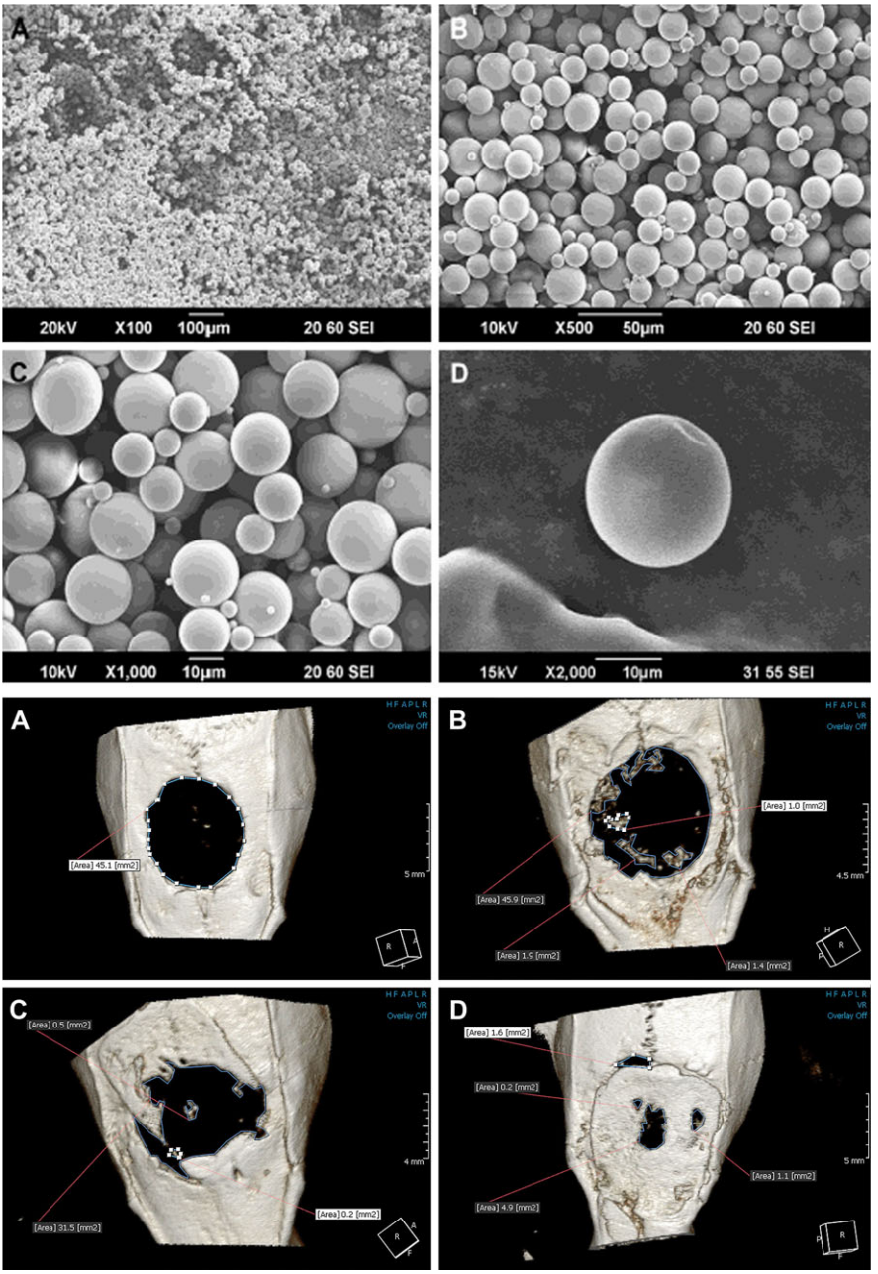


Fig. 7 *Left*: SEM images of the BMP-2-loaded microspheres. *Right*: Micro-CT images of rat cranial bone, at 11 weeks after implantation: (**A**) Negative control, (**B**) BMP-2-unloaded particulate leaching/gas foaming scaffold, (**C**) BMP-2-unloaded microstereolithography scaffold, (**D**) BMP-2-loaded microstereolithography scaffold [56]

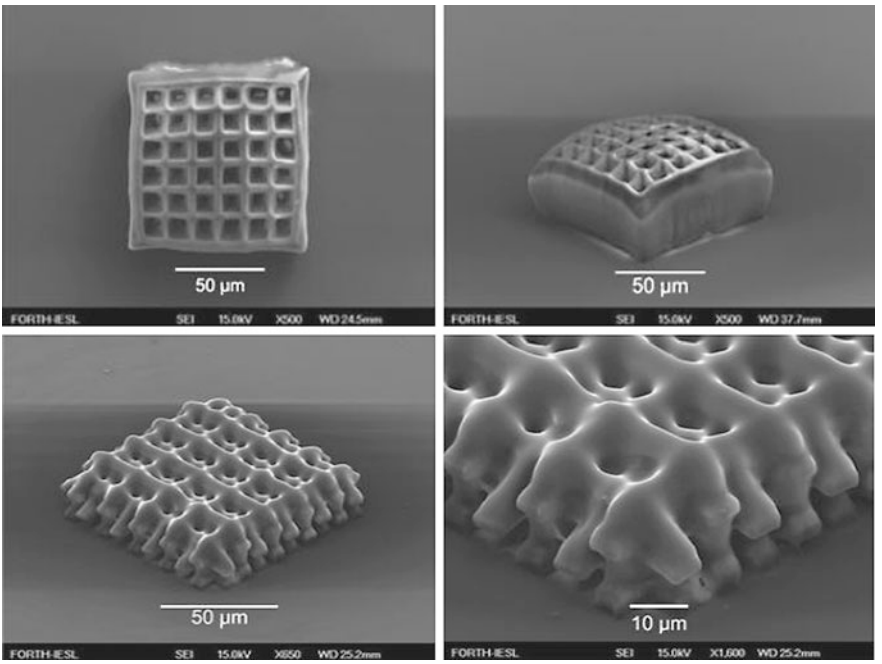


Fig. 8 SEM images of different 3D structures fabricated by two-photon polymerization [31]

microstructures, using methacrylated PCL-based oligomer (PCL-o) and Irgacure® 127 as UV photoinitiator. Live/dead staining analysis showed that hESC-derived neuronal cells, cultured on the PCL film for 7 days, were able to attach to the surface, though the material did not allow cell migration.

PLA is a biodegradable polymer available in three forms: L-PLA (PLLA), D-PLA (PDLA) and a racemic mixture of PDLLA [87]. The stereochemical structure of PLA can be modified by polymerizing a controlled mixture of L- or D-isomers, yielding amorphous or crystalline polymers [37]. PLA undergoes simple hydrolysis of the ester bond, and its degradation rate is highly dependent on the isomer ratio, the temperature of hydrolysis, as well the size and shape of the construct [37]. Recently, a photo-curable PDLLA-based material free of reactive diluents was developed, through functionalization with methacryloyl chloride [70]. This polymeric system was used for the fabrication of porous scaffolds with a gyroid architecture (Fig. 9), by using ethyl lactate as a non-reactive diluent. Pre-osteoblast cells, cultured on the scaffolds, showed good adhesion and proliferation. Jansen et al. [44] prepared biodegradable 3D porous scaffolds with a well-defined gyroid architecture (Fig. 9) and a porosity of 76 %, using photocross-linked networks based on fumaric acid monoethyl ester (FAME) end-functionalized PDLLA oligomers and N-vinyl-2-pyrrolidone, as a reactive diluent (PDLLA 3-FAME/NVP). Biological studies showed that mouse preosteoblast cells readily adhere and spread well onto disk-shaped polymeric networks. Koroleva et al. [52] used a photocurable and biodegrad-

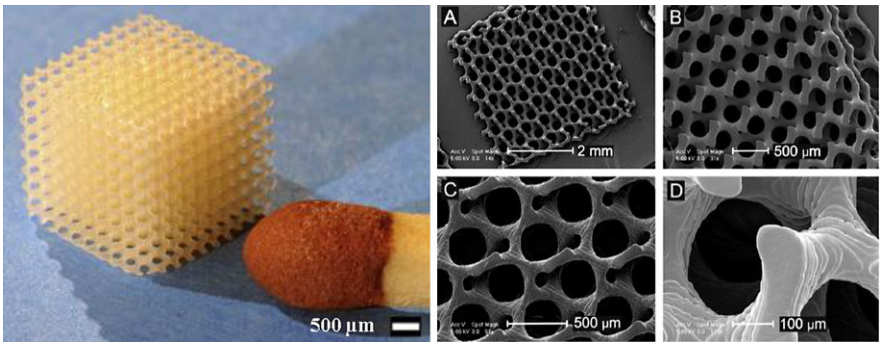


Fig. 9 *Left*: PDLLA scaffold built by stereolithography [70]. *Right*: SEM images of the PDLLA 3-FAME/NVP stereolithographic scaffold [44]

able PLA resin for the fabrication of 3D scaffolds. The potential of this polymeric material for neuronal tissue engineering applications was evaluated through the culture of rat primary Schwann cells and SH-SY5Y neuroblastoma cell line. Cells showed good adherence to the methacrylated PLA films, assuming spindle-like and flat cell morphologies when cultured on the 3D scaffolds. Results revealed the neurocompatibility of the developed constructs as well its ability to support cell proliferation.

Methacrylate end-functionalized PTMC macromers were used by Schüller-Ravoo et al. [89] to produce 3D constructs with a gyroid pore network. Before processing, the macromers were diluted using non-reactive propylene carbonate to decrease the viscosity and increase the processing temperature. The resulting network films exhibited high flexibility and elasticity, while the 3D porous scaffolds presented porosities in the range of 53–66 %. Meyer et al. [72] used the 2PP technique to produce 3D vessels with a branched tubular structure by, irradiating α, ω-polytetrahydrofuranether-diacrylate polymers. Tubular structures were obtained with a height of 160 µm, an inner diameter of 18 µm and a wall thickness of approximately 3 µm.

Synthetic hydrogels have also been processed through stereolithographic processes, such as poly(ethylene oxide) (PEO), poly(vinyl alcohol) (PVA), poly(butylene oxide) (PBO), poly(hydroxybutyrate) (PHB), polyacylamide, poly(hydroxypropyl methacrylamide) (PHPMA), poly(2-hydroxyethyl methacrylate) (PHEMA) and PEG [8].

PEG hydrogels are the most commonly used for conventional stereolithography [91], microstereolithography [63] and 2PP [80]. These materials exhibit high hydrophilicity, excellent biocompatibility, and can be functionalized with photoreactive end groups, such as acrylates or methacrylates, allowing the photopolymerization process [2]. Furthermore, PEG hydrogels can be susceptible to the hydrolytic degradation by the cleavage of the ester bonds, through either the introduction of proteolytically degradable peptide sequences into the backbone or by blending PEG with other biodegradable polymers [3, 24, 91]. Several works showed that PEG-based hydrogels can be modified with cell adhesive peptides, improving the cellular

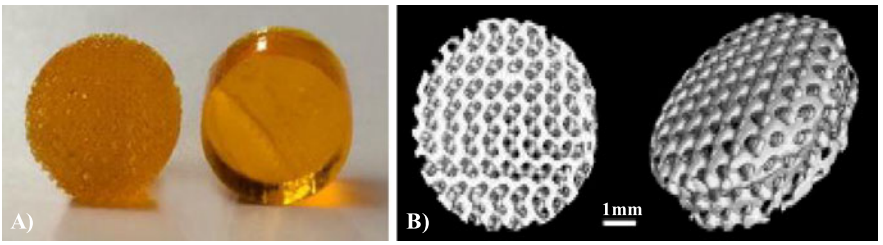


Fig. 10 (**A**) Porous (*left*) and solid (*right*) hydrogel structures built by stereolithography using methacrylate-functionalized poly(ethylene glycol)poly(D-Lactide), after several hours of extraction. (**B**) Micro-CT images of the porous constructs [91]

interaction, which can be processed by incorporating living cells at physiological conditions [24, 61–64, 115].

Seck et al. [91] produced porous and non-porous biodegradable hydrogel structures, using an aqueous photocurable polymer based on methacrylate-functionalized PEG/PDLLA macromers and Lucirin TPO-L as a visible light photo-initiator (Fig. 10A). Porous constructs were obtained with a gyroid pore network, 52 % of porosity and a fully interconnected pore network (Fig. 10B). Human mesenchymal stem cells, cultured on porous hydrogel structures, showed good adhesion and proliferation.

The fabrication of hydrogel scaffolds containing living cells and using stereolithographic processes was also tested. Bryant and Anseth [22] encapsulated chondrocytes into PEO hydrogels structures with a variable thickness, using an UV irradiation time of 10 minutes and a low light intensity ($\sim$10 mW/cm^2). The chondrocytes, encapsulated in the hydrogel structures and cultured *in vitro* during 6 weeks, remained viable and produced cartilaginous tissue. Lu et al. [63] used a dynamic masking system to produce scaffolds containing murine OP-9 marrow stromal cells. The cells were added to a polymeric system consisting of poly(ethylene glycol) diacrylate (PEGDA), dissolved into phosphate buffered saline and Irgacure 2959 as photo-initiator, and subsequently polymerized. After 24 hours of incubation, fluorescence microscopy analysis showed that cells maintained its viability. Using different fluorescently-labeled polystyrene microparticles, authors also showed the feasibility of the system to produce scaffolds with entrapped multiple biochemical factors presenting precise pre-designed and spatially-patterned layers (Fig. 11).

Lin et al. [61] incorporated living cells within PEGDA hydrogel scaffolds using a visible light-based projection stereolithography system. A commercial stereolithographic system, operating in a visible light mode (Hg illumination with UV barrier filter), was used to irradiate a monomer solution containing human adipose-derived stem cells, and lithium phenyl-2,4,6-trimethylbenzoylphosphinate as photoinitiator. After 7 days, it was possible to observe that cells maintained viability up to 90 %.

To reproduce the high complexity and heterogeneity of the human tissues, characterized by the presence of multiple ECM constituents and cells, it is critical to be able to produce 3D constructs containing heterogeneous layers, combining a variety

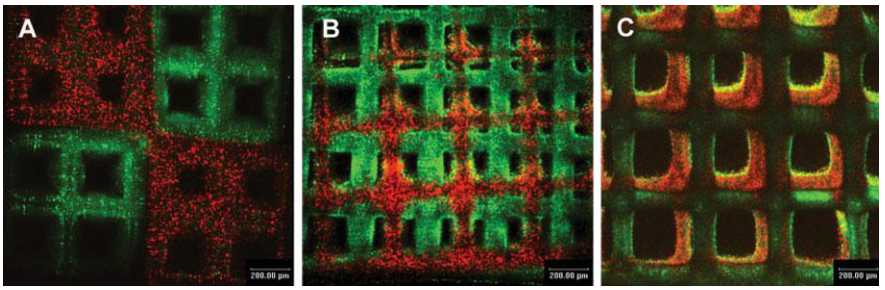


Fig. 11 Fluorescence confocal microscopy of PEGDA hydrogel scaffolds produced with pre-defined spatial-pattern in a single layer (**A**) and multi-layered scaffolds (**B**, **C**) containing either FITC or Cy5-labeled polystyrene particles [63]

of biomaterials and cells with precise spatial arrangement. Arcaute et al. [2–4] explored stereolithography for fabricating multi-material spatially controlled bioactive scaffolds. To accomplish multi-material fabrication, a mini-vat setup was designed, allowing for self-aligning X–Y registration during fabrication. The mini-vat setup allowed the construct to be easily removed and rinsed, and different photocrosslinkable solutions to be easily removed and added to the vat. Multi-material scaffolds were fabricated by including controlled concentrations of fluorescently labeled dextran, fluorescently labeled bioactive PEG or bioactive PEG in different regions of the scaffold (Fig. 12). Human dermal fibroblast cells were seeded on top of the fabricated scaffolds. Spatial control was successfully showed in features down to 500 μm.

Ovsianikov et al. [81] combined the 2PP technique and the laser-induced forward transfer (LIFT) process to produce 3D multicellular tissue constructs. The 2PP technique was employed for the fabrication of PEGDA scaffolds, which were subsequently seeded with vascular smooth muscle-like cells (within the outer scaffold area) and endothelial cells (into the inner scaffold area), using the LIFT process. To allow the fabrication of 3D constructs containing multiple hydrogel compositions and cell types with control over the spatial distribution of cells and bioactive molecules, Chan et al. [24] modified a conventional stereolithographic apparatus. These authors showed the production of PEGDA constructs with encapsulated/3T3 cells through the manual addition of individual layers of cell-containing photopolymer, preventing the cells settling to the bottom of the stereolithographic vat due to gravity.

Zorlutuna et al. [115] used stereolithography to produce spatially organized 3D co-culture of multiple cell types to investigate cell-cell interaction and the microenvironments of complex tissues. Two layer-constructs, using different cell types and hydrogels, were produced as shown in Fig. 13. The first layer was produced by polymerizing poly(ethylene glycol) methyl ether methacrylate (PEGMA3400) containing adipose-derived stem cells (ASCs), while the second layer contains primary hippocampus neurons (HNs) and skeletal muscle myoblast (MCs) cells encapsulated in oxidized methacrylic alginate and poly(ethylene glycol) methyl ether methacrylate 1100 (OMA-PEGMA1100).

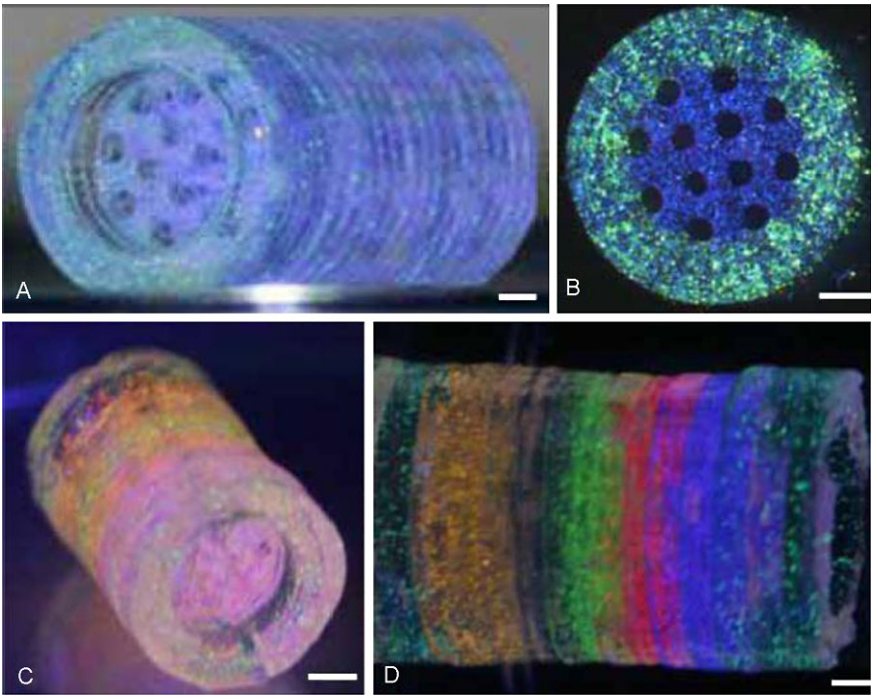


Fig. 12 Nerve guidance conduit produced using different colors of fluorescent particles in the PEG-based solution (the scale bar is 1 mm) [4]

3.1.2 Natural Polymers and Hydrogels

Recently, the use of natural polymers for biomedical applications has attracted an increase interest, due to its excellent properties, such as biodegradability, biocompatibility, low toxicity, availability, similar properties to the human tissues and inherent cellular interaction [42, 84]. When compared to synthetic polymers, these materials offer several advantages including biological signaling, cell adhesion and cell responsive degradation [87]. The main drawbacks of natural polymers regards the limited stability of its mechanical properties, low processing window and the risk of immune rejection [42, 86, 87, 96]. The use of these materials on stereolithography is still in its infancy, despite recent advances on the modification of natural polymers to allow its photo-crosslinking. Generally, natural polymers are modified by the introduction of methacrylate groups on the backbone to yield photopolymerizable materials. Examples include photocrosslinkable natural polymers like gelatin [76, 90], hyaluronic acid [5], alginate [45], chitosan [114] and silk fibroin [107]. These materials are explored in stereolithography manly to produce hydrogel scaffolds with and without encapsulated cells.

Gelatin is a natural polymer derived from collagen, one of the main constituents of ECM matrix, commonly used to produce scaffolds for tissue engineering, due

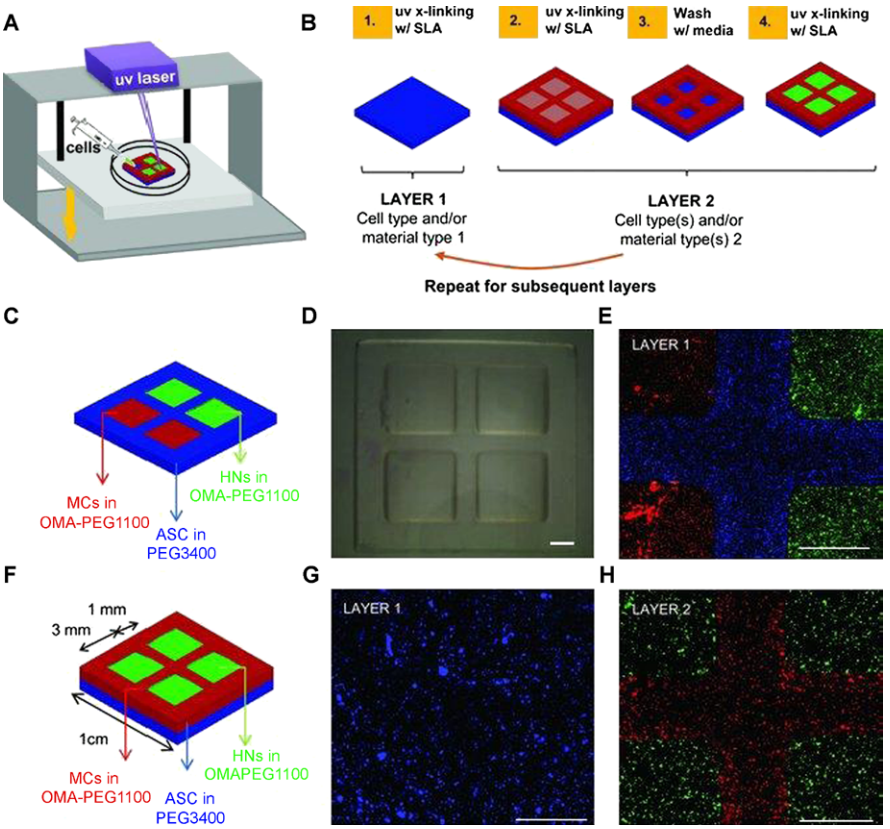


Fig. 13 Illustration of SLA process (**A**). Fabrication sequence (**B**). CAD models (**C**, **F**). Stereomicroscopy image of the produced hydrogel (**D**). Fluorescence microscopy images of the encapsulated MCs (*red*), HNs (*green*), and ASCs (*blue*) cells in different regions of the same layer (**E**); ASCs in the first layer (**G**) and HNs and MCs in the second layer (**H**). Scale bar represents 1 mm [115]

to its biodegradability and ability to provide good cell interaction and proliferation. Gauvin et al. [38] used a projection stereolithography system to produce scaffolds using gelatin methacrylate. Human umbilical vein endothelial cells, cultured on the porous scaffolds, exhibited good proliferation and maintained the endothelial phenotype. The 2PP technique was employed by Ovsianikov et al. [79] to produce 3D scaffolds using methacrylamide-modified gelatin (GelMOD), and Irgacure 2959 as photoinitiator (Fig. 14). After gelatin modification with the methacrylamide, it was observed that the *in vitro* degradation behavior of the polymer was not affected. The gelatin-based hydrogels showed ability to support the adhesion and proliferation of porcine mesenchymal stem cells, as well the capacity to induce the cellular differentiation into the anticipated lineage, upon applying osteogenic stimulation.

Basu et al. [19] prepared collagen (type I, II, IV) scaffolds with benzophenone dimer photoactivator using multiphoton (two- and three-photon excitation) poly-

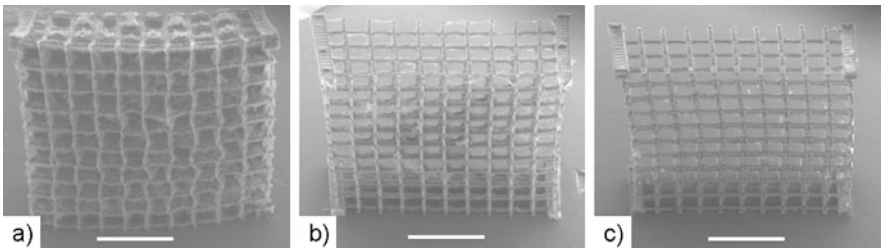


Fig. 14 Gelatin-based scaffolds built by 2PP process. Bar corresponds to 1 mm [79]

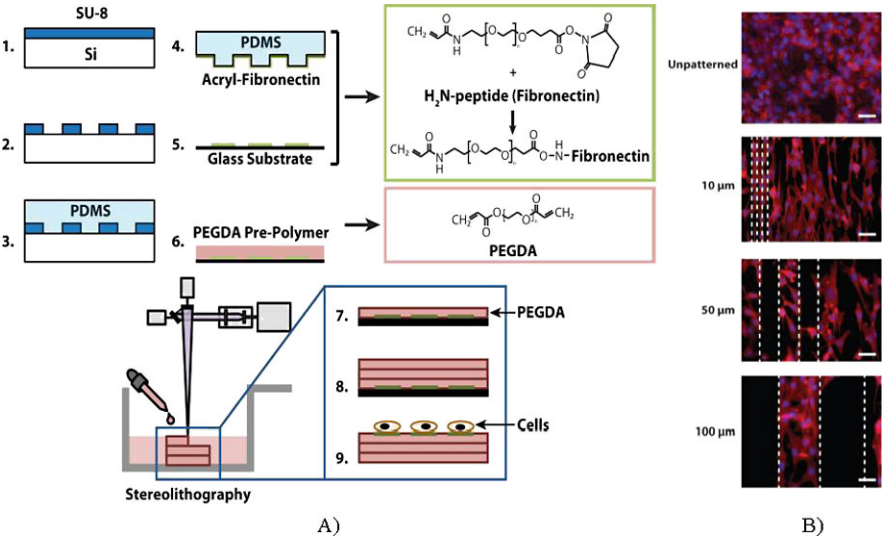


Fig. 15 (**A**) Fabrication procedure used to create patterns of acryl-fibronectin on the backbone of PEGDA hydrogels produced by stereolithography. (**B**) Fibroblasts aligned on the unpatterned and patterned hydrogels substrates [23]

merization. Scaffolds were crosslinked with the benzophenone dimer, and structures at sub-micrometer and micrometer scale were obtained. Results showed that the degradation rate of these structures can be tailored by changing the degree of cross-linking of collagen during fabrication. Primary human dermal fibroblasts cultured on the collagen structures showed good adhesion, so there is biocompatibility in the designed constructs. Chan et al. [23] developed a simple method for aligning cells on 3D hydrogels, by combining the micro-contact printing technique and stereolithography (Fig. 15). They modified fibronectin with acrylate groups, creating patterns of acryl-fibronectin on the backbone of PEGDA hydrogels produced by stereolithography, using a micro-contact printing technique. The cell alignment in the direction of the patterns created on the hydrogel substrates, was showed by the culture of the NIH/3T3 mouse embryonic fibroblasts.

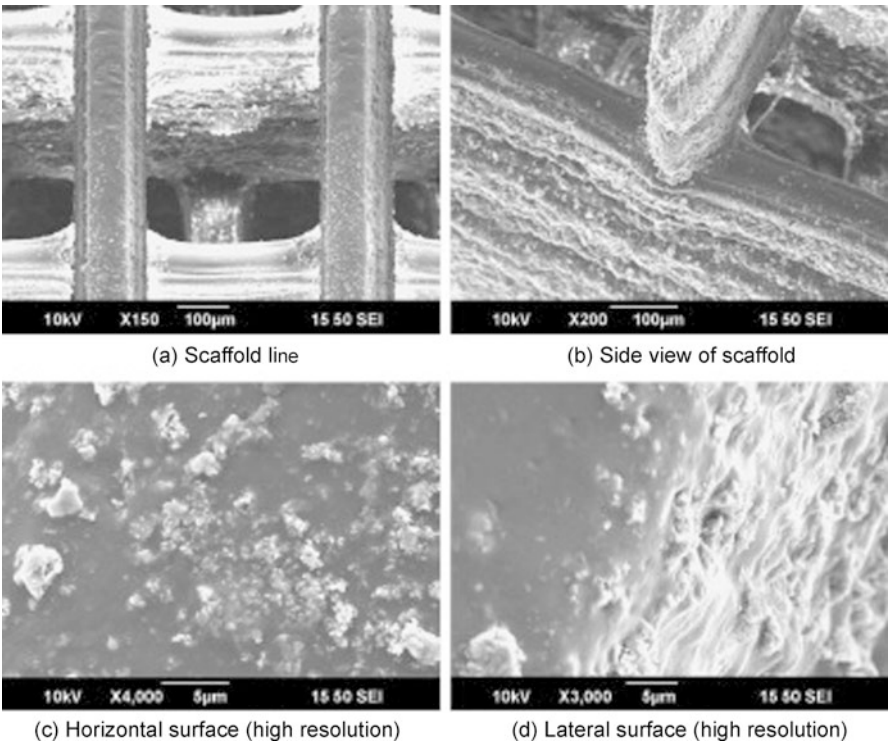


Fig. 16 SEM images of the 3D PPF/DEF scaffolds containing hydroxyapatite [55]

3.2 Polymer/Ceramic Blends

Composite systems, based on polymer and ceramic materials, offer the possibility to produce constructs with increased mechanical properties and bioactivity [71, 87]. To produce polymer-ceramic scaffolds through stereolithography, ceramic particles (hydroxyapatite (HA) and tricalcium phosphate (TCP)) are homogenously dispersed on the polymeric material and subsequently photopolymerized [20, 30, 55, 60]. These scaffolds are usually stiffer and stronger than the polymeric ones, as a result of the incorporation of powder particles. However, the powder not only affects the mechanical properties of the scaffolds, but also increases the viscosity of the polymeric system, affecting the solidification process and the light penetration depth [20, 71].

Lee et al. [55] produced PPF/DEF scaffolds containing 7 % of HA nanopowder, and BAPO as a photoinitiator using the microstereolithography process. MC3T3-E1 pre-osteoblasts, cultured on these scaffolds during 2 weeks, showed better adhesion and proliferation when compared with scaffolds without HA (Fig. 16).

Sharifi et al. [93] produced soft nanocomposite hydrogel structures by the polymerization of a solution, consisting of methacrylate-functionalized triblock copolymers of PEG and PTMC with colloidal dispersions of clay nanoparticles (Laponite

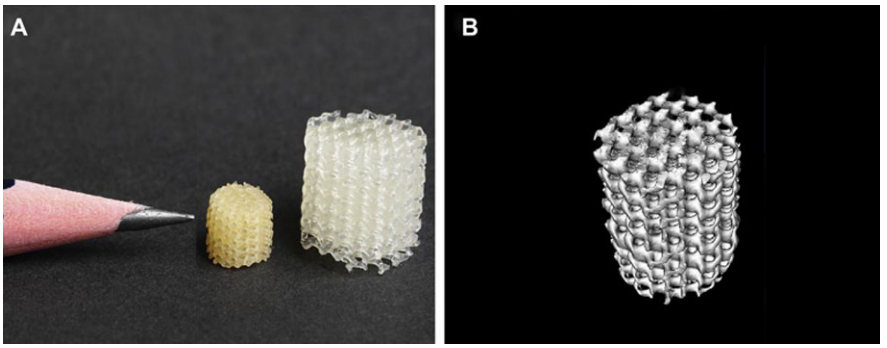


Fig. 17 (**A**) Macroscopic image of the structures built by stereolithography in the equilibrium water swollen state and after drying. (**B**) Micro-CT image of the structure in water swollen state [93]

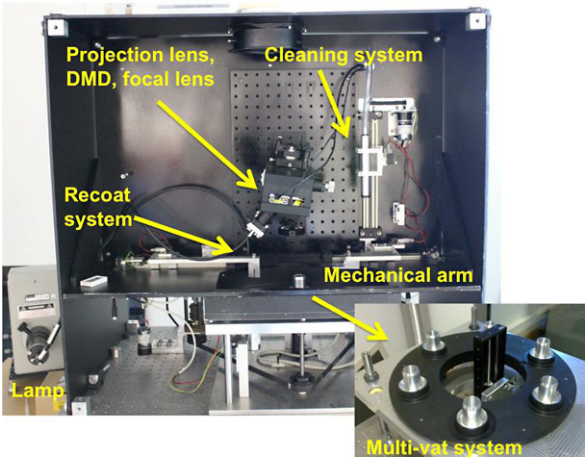


Fig. 18
Stereo-thermal-lithography system

XLG) at different concentrations (2.5 and 5 wt %). Unconfined compression tests were performed regarding the swollen hydrogels, showing that an increasing on the concentration of the Laponite nanoclay improved the compressive modulus. Scaffolds with a gyroid pore network architecture, obtained by stereolithography, showed an interconnected pore network (Fig. 17). In another work, bioactive glass S53P4, an inorganic material with the ability to interact with bone tissue, was combined with a methacrylated PCL polymer to produce porous scaffolds [34]. Bioactive glass was homogeneously distributed through the scaffold and its surface, improving the compression modulus of the construct and the cellular activity of the seeded fibroblasts.

Figure 18 illustrates a mask-based multi-photon and multi-material stereolithographic system. This system, called Stereo-thermal-lithography (STLG), uses a mercury lamp of 350 W as a light source. Appropriated filters split the radiation into two different wavelengths: UV radiation and near-infrared (near-IR) radiation. Optical

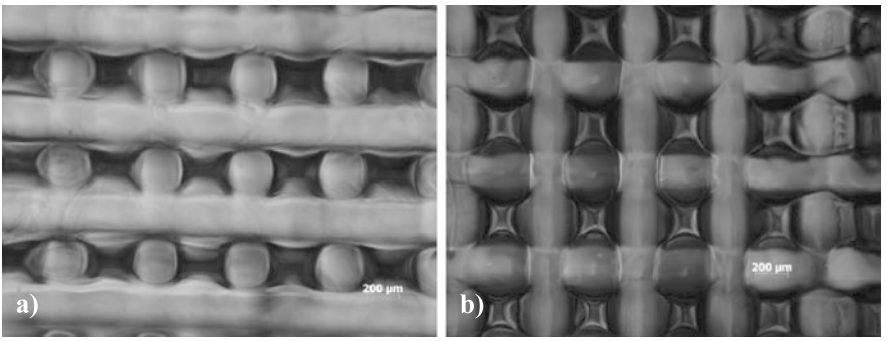


Fig. 19 (**a**) Poly(HEMA) scaffolds containing 2 wt % initiator. (**b**) Poly(HEMA) scaffolds containing 1 wt % initiator

fibres, projection and focal lenses irradiate a UV-DMD and a IR-DMD. A dichroic mirror captures the images projected on both DMDs (1024 × 768 pixels, 14 mm in size), combining them into a single image that is transferred to the liquid polymer. The equipment also includes a multi-vat system enabling the fabrication of multi-material constructs. The vertical displacement of the platform is secured by a positioner uniaxial MYCOSIS® Translation Stage VT-80. This positioner allows vertical increments of 1 µm, at a speed ranging between 0.001 and 20 mm/s.

STLG intends to solve some of the major limitations of conventional stereolithography, such as efficiency, speed and accuracy. Its main advantages over other stereolithographic processes are [10] as follows:

- a more efficient generation of radicals due to the combination of multiple light effects;
- use of small concentrations of the two types of initiator (UV and near-IR initiators), enabling the radiation to penetrate deeper into the polymer;
- a more localized curing reaction, improving the accuracy of the produced models;
- the system has more tunability. Three subsystems can be considered. ***Subsystem A***: uses UV radiation to solidify a photopolymer containing a certain amount of UV initiator. ***Subsystem B***: uses IR radiation to solidify a photopolymer containing a certain amount of IR initiator. ***Subsystem C***: uses both UV and IR radiation to solidify a photopolymer incorporating a certain amount of UV and IR initiators.

The STLG process has been used to produce highly reinforced polymeric systems with metallic and ceramic constructs, PVA and polyHEMA scaffolds (Fig. 19) [33]. The effect of light intensity, photo-initiator concentration, metallic/ceramic powder concentration and powder particle size has been investigated.

Stereolithography has also been used for the direct fabrication of ceramic scaffolds, where the ceramic particles are dispersed on the photocurable polymer and subsequently polymerized. Upon polymerization, the polymer (binder material) is removed using an appropriated thermal or dissolution treatment, and the ceramic particles sintered in order to confer the final properties to the construct [20, 82].

Levy et al. [60] used a ceramic suspension consisting of HA powder and photocurable resin to produce ceramic scaffolds for orbital floor prosthesis. After polymerization, the resin was removed by heating and the resulting ceramic scaffold consolidated by sintering. By combining stereolithography and gel casting, Bian et al. [20] produced an osteochondral beta-tricalcium phosphate/collagen scaffold containing a bone phase with a 3D channel network composed of β-TCP, a cartilage phase consisting of collagen and a transitional interface between the bone and cartilage. Stereolithography was used to produce both bone and transitional phase, through the polymerization of a ceramic suspension. After processing, the binder was removed by pyrolysis, and the porous structure sintered. To produce the cartilage phase, a solution of type-I collagen was casted into a cylindrical mould on the surface of the ceramic scaffold, and then freeze-dried. Finally, the scaffold was immersed into a glutaraldehyde solution (0.5 wt %) to allow the cross-linking of collagen. Bone marrow stromal cells, cultured on the porous scaffolds under perfusion attached well, covering about 60 % of the structure after 7 days.

Ceramic scaffolds can also be obtained through an indirect approach that uses stereolithography to produce lost-moulds, in which ceramic suspensions are casted. The mould is removed by pyrolysis and the scaffold submitted to a sintering process [8]. Chu et al. [30] produced hydroxyapatite scaffolds with two internal architectures (radial and orthogonal design), by casting HA/acrylate suspensions into epoxy moulds created by stereolithography. After polymerization, the epoxy mould and the acrylic binders were removed by sintering, and the scaffolds sintered in a furnace. Scaffolds were implanted in the mandibles of Yucatan minipigs during 5 and 9 weeks. Results showed that the scaffold design has a great influence on the bone regeneration. Scaffolds with a radial design induced the regeneration of new bone tissue as an intact piece at the center of the construct, while the scaffolds with an orthogonal design induced the bone formation as an interpenetrating matrix with the HA implant. A similar approach was used by Kim et al. [49] to produce hydroxyapatite scaffolds, using lost-moulds fabricated by microstereolithography. Recently, Chopra et al. [29] produced glass-ceramic (apatite–mullite glass-ceramic, LG112) scaffolds with simple cubic structure, by gel-casting into moulds produced by stereolithography (Fig. 20a). The moulds were produced by the polymerization of an acrylate polymer and removed prior sintering. Primary human osteoblasts, cultured on the scaffolds composed of only a few slices (Fig. 20b) and square channels of width 400 or 600 μm, showed good adhesion, spreading and proliferation. Despite cells readily proliferate on the scaffold surface, its penetration into the structure was limited, as observed by confocal microscopy (Fig. 20c–e).

4 Final Remarks

Stereolithographic processes have been widely used in tissue engineering for the fabrication of scaffolds for the regeneration of different tissues (e.g. bone, skin, neurons), due to its accuracy, precision, resolution and ability to process a large

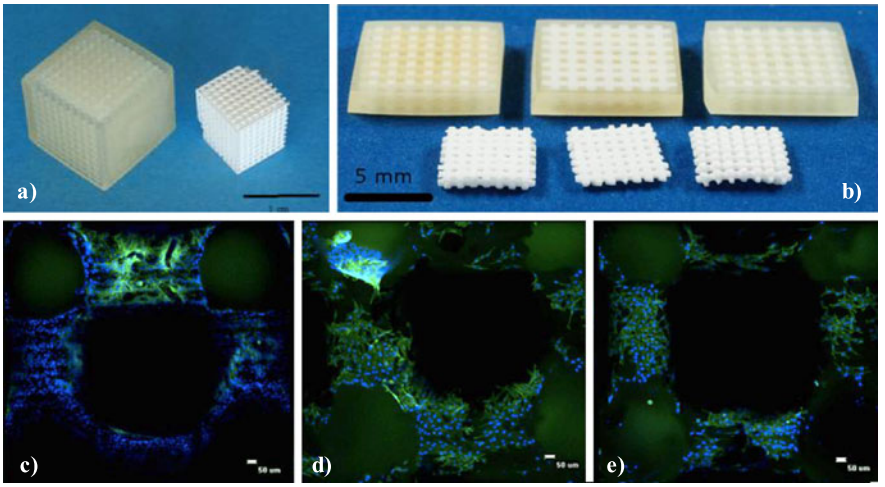


Fig. 20 (**a**) Mould built by stereolithography and respective sintered glass-ceramic scaffold obtained by gel-casting. (**b**) Scaffolds used in the cell culture studies. Confocal fluorescence micrographs of osteoblasts, stained with FITC-conjugated phalloidin for f-actin and DAPI for nuclei on the *top* (**c**), *middle* (**d**) and *bottom* (**e**) of the scaffold, with square channels of with 400 μm [29]

variety of polymer and ceramic materials at physiological temperatures in the presence of cells and growth factors. In spite of current advances, there is a need to produce photocurable systems exhibiting appropriate biocompatibility, biodegradability and cellular interaction to allow the fabrication of scaffolds mimicking the natural ECM constitution and organization. Recent works showed the ability of stereolithography to produce 3D environments for studying the interactions between different cell types, and induce the cell alignment. However, the fabrication of clinically relevant 3D constructs incorporating different cell types in an organized manner is still a great challenge. In order to improve the clinical application of tissue-engineered products obtained by stereolithography, there are relevant challenges to be addressed:

- The development of advanced photocrosslinkable hydrogels mimicking the ECM constituents, allowing the incorporation of multiple cell types with precise and controlled distribution;
- The heterogeneity of natural tissues requiring the fabrication of multimaterial scaffolds resembling its structural and biological organization, providing an optimal environment for the cell attachment, proliferation, differentiation and growth;
- The development of parallel processing strategies to increase the processing rate of stereolithographic techniques and improve the large-scale production towards clinical applications;
- The fabrication of hydrogel scaffolds incorporating living cells requiring the development of advanced optical imaging techniques, capable to perform analysis within the gels without affecting the cell viability or destroying the gels;

- Today, research works are mainly investigating the effect of stereolithography on the cell adhesion, proliferation and differentiation. There is a need for works on the influence of the processing parameters, such as the light intensity and the exposure time on the phenotype of the cells.

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